



Deep learning based real-time Industrial framework for rotten and fresh fruit detection using semantic segmentation

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Received: 6 November 2020 / Accepted: 16 November 2020
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Abstract

Computer vision finds wide range of applications in fruit processing industries, allowing the tasks to be done with automation. Classification of fruit's quality and thereby gradation of the same is very important for the industry manufacture unit for production of best quality finished food products and the finest quality of the raw fruits to be sellable in the market. In the present paper, detection of rotten or fresh apple has been accomplished based on the defects present on the peel of the fruit. The work proposes a semantic segmentation of the rotten portion present in the apple's RGB image based on deep learning architecture. UNet and a modified version of it, the Enhanced UNet (En-UNet) are implemented for segmentation yielding promising results. The proposed En-UNet model generated enhanced outputs than UNet with training and validation accuracies of 97.46% and 97.54% respectively while UNet as the base architecture attaining an accuracy of 95.36%. The best mean IoU score under a threshold of 0.95 attained by En-UNet is 0.866 while that of UNet is 0.66. The experimental results show that the proposed model is a better one to be used for segmentation, detection and categorization of the rotten or fresh apples in real time.

1 Introduction

Artificial intelligence (AI) with the aid of computer vision is boosting various sectors for quality production with high efficiency. In fields of agriculture and food industry, AI extends its help to the farmers and the manufacturers to improve their effectiveness and to overcome the traditional challenges under environmental hostile impacts. The adoption of AI in the agro-based industries has strengthened the technology to a greater extent. With the implementation of automation techniques, the food processing units have shown promising outcomes owing to excellent production and smart packaging.

In the twenty-first century, both fruit and food processing industries are undergoing soaring competitive positions. Global trade and market flow of fruits and vegetables determines geographical closeness between exporter and importers. In case of exporting or importing, there is a long and time-consuming process of transportation which causes hindrance in checking the quality of rotten or nearly rotten fruits amongst a bulk quantity of fruits. Thus production of fruits is likely to shrink to a greater extent as compared to previous years' world fruit production and trade. Besides all other obstacles, uncertain weather conditions, climate change and temperature rise are other major causes of concern behind the declination of trade. Moreover, apart from export and import of fresh fruits, the food processing sectors is also severely hampered owing to scrutinization of rotten fruit and degradation in its quality. Thus a professional and efficient method is required for sorting and gradation of good quality fruits. Herein application of computer vision based systems facilitated with the image processing techniques yields highly accurate and precise results. The methodological proficiency enables computer vision based systems for checking defective fruits and thereby improving the potential for healthy fruit production. Computer vision makes possible, practices for monitoring, harvesting and managing effectively various

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types of fruits in the orchards and crops in the fields (Ni et al. 2020a). Quality inspection and the categorization of fruits in the traditional methods require skilled persons. Human involvement in performing the task of detecting rotten or defective fruits varies from person to person based on just seeing and sensing from outside. Furthermore, the requirement for enhanced quality food products is increasing under new governmental regulations and consumer market demands. An automated system based on capturing the image of the fruit thereby processing on the detailed features of the image results in high precision and fast response. Computer vision systems supported with deep learning architectures for detection of rotten or fresh fruit are becoming popular among the researchers (Kang and Chen 2020). These types of automated systems generate accurate outcomes.

The present work portrays the idea of a computer vision based system for detection of rotten or fresh apple. The rotten apple is identified using segmentation technique wherein the rotten portion of the fruit is segmented out. The segmentation process has been done based on deep learning architecture, taking RGB image of apple (rotten/fresh) as input to the system. Color being the primary identifier by appearance determines the quality of the fruit. As every apple is different and are not identical, the conventional image processing techniques are not going to help out with better results. In the last 5–6 years with increased computation power and availability of the dataset, the deep convolution networks performed excessively well in visual recognition and identification.

In the present paper, the segmentation task is done using UNet and a modified version of it, the Enhanced UNet (En-UNet). UNet has been widely used for diagnosing medical images and the network outperformed the traditional segmentation techniques (Ronneberger et al. 2015). UNet or any of its modified form has not been implemented so far in horticulture for the purpose of segmentation in the fruit's image. The En-UNet has been applied and better results are obtained than the original UNet in terms of accuracy, precision and other model performance parameters. UNet architecture is based on the concept of encoder and decoder. The encoder comprises of the traditional layers of convolution and maxpooling. Following the encoder, the decoder is a symmetric path performing expansion using transpose convolution layers for precise localization. In En-UNet, the depth is increased by adding extra layers of convolution, maxpooling and transpose convolution. Around 4035 RGB images of apples are used for training and testing the architecture containing both fresh and rotten apples images in the dataset. The results obtained are satisfactory with the proposed model and hence can be implemented with a developed automated system in the domain of horticulture.

The remaining paper is structured as follows. Section 2 represents the related work of fruit segmentation techniques while Sect. 3 highlights the methodology and the model description. The experimental results and discussion are depicted in Sect. 4 and the entire work is concluded in Sect. 5.

2 Related works

Segmentation techniques for fruit detection, identification and classification have been studied extensively in the past decades. A deep learning segmentation has been implemented in analyzing the blueberry fruit traits like cluster compactness, maturity of the fruit and number of berries for ease of the blueberry breeders (Ni et al. 2020b). A masked R-CNN model has been trained and tested for the detection purpose of the blueberries on maturity. The model resulted with average precision for validation and test data is 78.3% and 71.6% considering 0.5 over union threshold. The accuracies are 90.6% and 90.4% respectively. Green Shoot thinning (Majeed et al. 2020) in wine grapes is done for quality wine with an automated process that would yield higher efficiency and good performance. Deep learning architectures like Segnet (Badrinarayanan et al. 2017) and FCN are implemented for the successful semantic segmentation. The FCN model viz. FCN VGG-16 achieved better F1 score as compared to Segnet-VGG16, Segnet-VGG19 and FCN-Alexnet. In (Kang and Chen (2019)), the authors proposed a network model for performing real-time detection with segmentation of apples and branches. The developed network mainly utilized the atrous spatial pyramid pooling and gate feature pyramid arrangement for feature extraction. ResNet-101 with 0.832 F1 score showed very high performance in detection of the apples with accuracy of 87.6% on segmentation task. A voxel-based convolutional network (VCNN) (Jin et al. 2019) is implemented for classification and segmentation of structural components in maize. The comparison results with some traditional clustering techniques and deep learning methods (Pointnet and Pointnet++) showed very good results. The Lidar's implementation for the separation of structural components in maize by means of classification and segmentation paved a new path to be applied for other fields as well. Detection of foreign objects (eg. dried leaves, paper, packing materials, plastics or metal parts) in walnuts has been done using fully convolutional deep network (Rong et al. 2019). The proposed model acknowledged with the segmentation task generating good outputs. Kumari et al. (Kumari et al. 2020) in their paper has proposed a hybrid model for classification of mangoes using segmentation technique. The approach is based on BPNN and discriminant analyzer. An enhanced

method using Fuzzy C-means clustering has been implemented for the segmentation and back propagation based discriminant classifier is used in the classification practice. For the process of feature selection Maximally Correlated Principal Component Analysis (MCPCA) is utilized. Authors in the paper (Liu et al. 2019) have undergone the segmentation task for cucumber detection. Masked region-based convolution (RCNN) with FPN (Feature Pyramid Network) has been executed. For identification purpose, deep learning architectures have excelled and generated perfect outputs. Identification of flowers (apple, peach and pear) by the proposed method (Dias et al. 2018) generated semantic segmentation. An end-to-end residual CNN has been employed for the work. Segnet, an architecture effective for semantic segmentation, is a fully convolutional network with the deployment of encoder and decoder. The network have been compared with FCN (Long et al. 2015), Deeplab-large FOV (Chen et al. 2017), DeconvNet (Noh et al. 2015) and reported to have better segmentation results.

The traditional machine learning approaches have put forward many detection, segmentation and classification algorithms. However the classical methods involve a lot of hand crafted procedures for feature selection and extraction. Yang et al. (2020) in their paper suggested a method of colorful fruit image segmentation taking texture features into account. The process of segmentation followed by classification using multi-class support vector machine (MSVM) can be used for multi-class classification (Shafi et al. 2018). Algorithms such as Naïve Bayes, ANN and decision Tree have also been applied for sorting of fruits like orange (Wajid et al. 2018). Clustering technique viz. K-Means clustering have been applied for detecting the rotten part of defective apples (Mhapne et al. 2019). Different classical segmentation techniques viz. color, edge and marker segmentations generates useful results in detection of rotten portion in vegetables like tomatoes (Roy et al. 2019a). Mask generation (Roy et al. 2019b) for segmentation have been an efficient way for producing desired segmented outputs. Deep learning approach introduces mask R-CNN (Yu et al. 2019) which involves two processes for segmentation–detection of component and generation of mask.

The present work of segmenting the rotten portion of apple has been done with UNet as the backbone architecture along with some modification in the network for enhanced output. UNet has been framed with the strategy of data augmentation. The model comprises of contracting path for capturing context followed by symmetric expansion path for acute localization. The architecture is very fast in context of processing. For an image of size 512×512 a recent GPU takes less than a second for the segmentation purpose. Unet finds wide range of

applications in medical imaging for detection and identification functions. Segmentation of pancreas (Li et al. (2019)) for the diagnosis of cancer is one such instance of UNet's application in medical domain. UNet with different modifications in its architecture have been adopted and came up with new names with the variations. Zhou et. al (2018) proposed UNet ++ (a nested UNet), fundamentally designed keeping encoder and decoder blocks with dense skip pathways in nested fashion. RIC-UNet (Zeng et al. 2019), AD-UNet (Luo et al. 2019), FD-UNet (Guan et al. 2019) are the other modified versions of the UNet architecture designed for segmentation of nuclei in histological images, in fundus images or artifacts removal in 2D PAT images. The mentioned UNet or some of the deep learning architectures have been applied universally for medical imaging analysis. However the architectures performing segmentation (Chen et al. 2019) for the identification (Häni et al. 2020) of fruits, vegetables, detection of defective vegetable or fruit and recognition of anomalies (Nardari et al. (2018)) in the field of horticulture, can function efficiently with profoundness. Though many segmentation algorithms have already been used in recent past but for segmentation of rotten or fresh apple, UNet or any of its modified versions is implemented for the first time for such application.

3 Materials and method

The overall methodology of the segmentation procedure is shown in Fig. 1. The proposed method is initialized with the required dataset ([www.kaggle.com, sriramr, fruits-fresh-and-rotten-for-classification](http://www.kaggle.com/sriramr/fruits-fresh-and-rotten-for-classification)) to the training model. The data set comprises of fresh and rotten apple RGB images. The dataset is preprocessed using image processing and are fed to the training module with a continuous monitoring on the training accuracy and loss. In the workflow diagram followed by the training module, the testing module validates the accuracy of an unknown image (fresh/rotten) and generates the required output based on the type of the input image. As the task is activated for segmenting out the rotten portion of an apple's image, hence if any rotten apple's image is fed as input, the segmented portion of the rotten part is achieved at the output of the testing module. The descriptions of the different functional blocks present in the workflow diagram are provided below:

3.1 Data Preprocessing

The input RGB images are preprocessed and are converted into gray images. Further, the gray images are masked by the means of thresholding and inverse binarization as

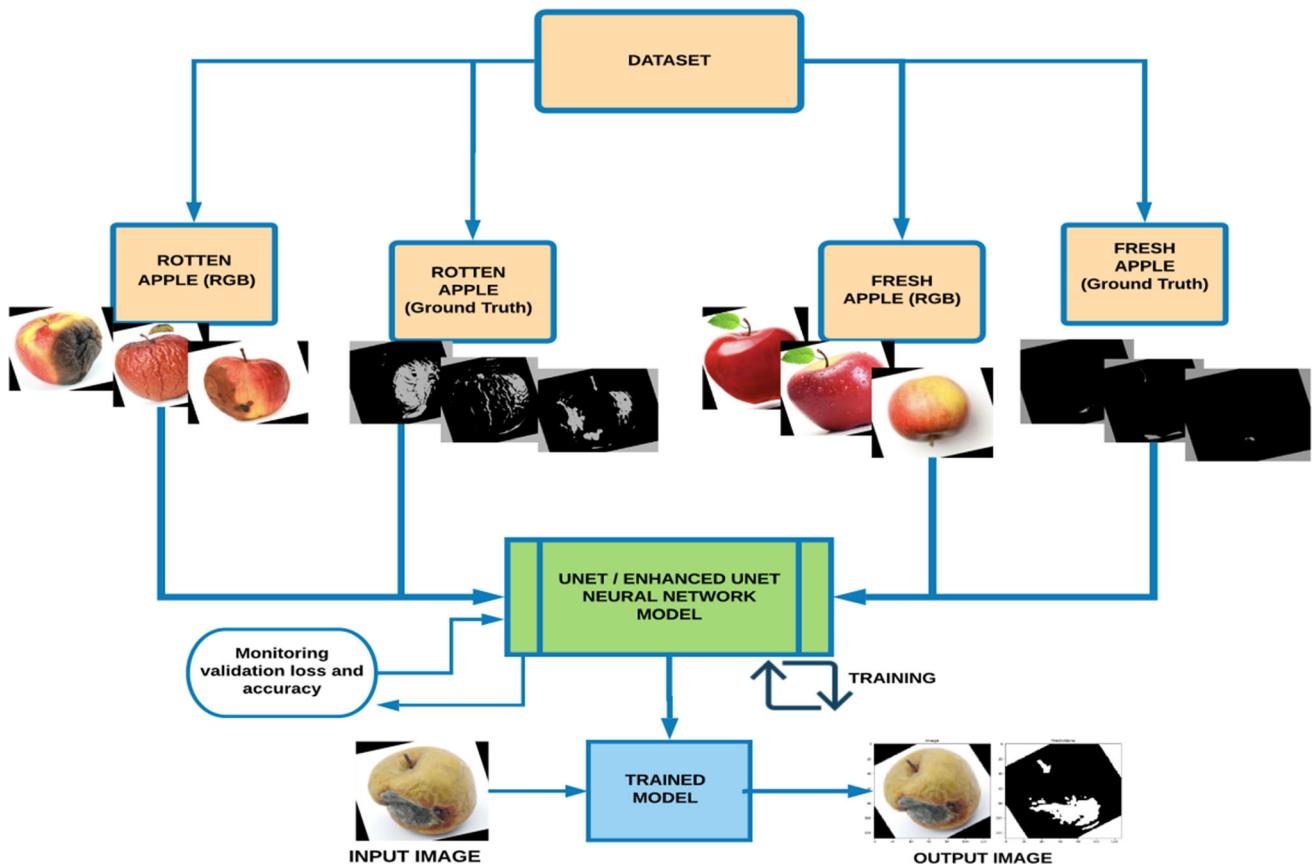


Fig. 1 Workflow diagram of proposed methodology

shown in Fig. 2. After this step the binary masks of the input RGB images are obtained. In the masked output binary image only the rotten portion is present (white color), for a rotten apple's RGB image and the mask remains completely black for a fresh apple. The approach of generating masks has been done to produce a general

dataset which reflects the ground truth of the input image distinguishing the rotten and fresh apple as shown in Fig. 3. The set of masks produced both for the fresh and rotten apples are fed to the training module of the automated system for segmenting out the rotten part in any input RGB image of an apple.

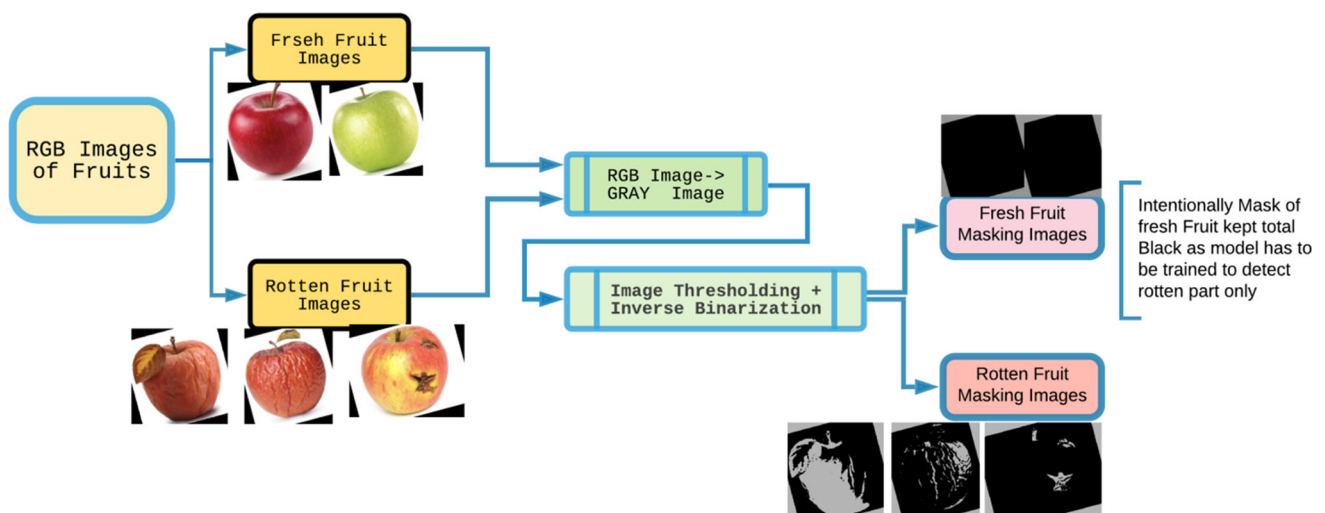


Fig. 2 Data preprocessing and generation of ground truth images (binary masks)

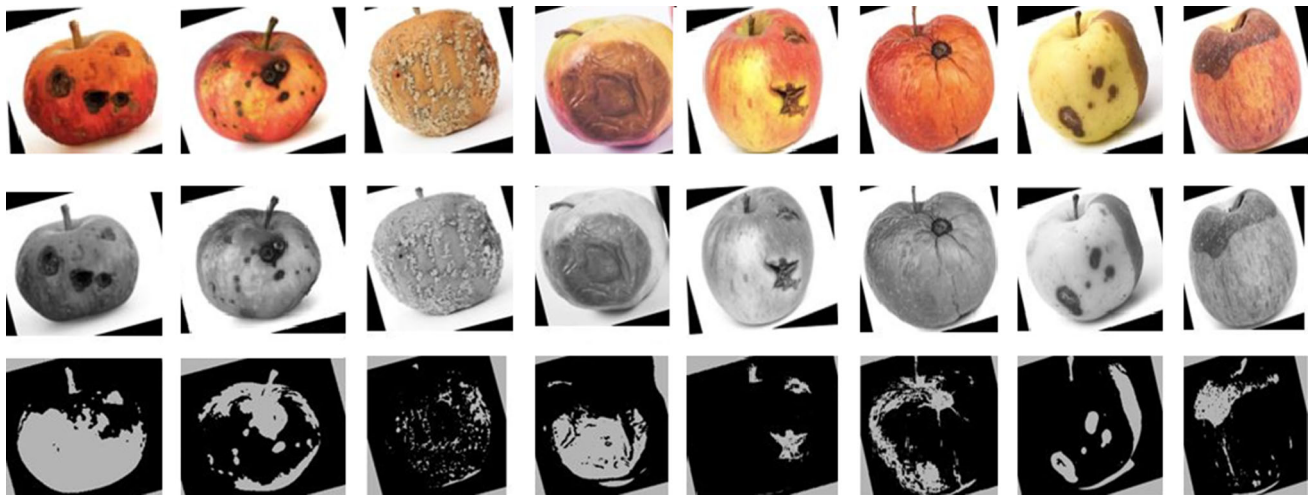


Fig. 3 Conversion of RGB to Gray images (Apples) and the corresponding binary masks

3.2 Data set

The fruit image data set ([www.kaggle.com, sriramr, fruits-fresh-and-rotten-for-classification](http://www.kaggle.com/sriramr/fruits-fresh-and-rotten-for-classification)) is used in the present work for the segmentation purpose. The dataset contains many categories of fruits but in this case only apple has been taken for the mentioned task. The apple's data is RGB image data, containing 1693 fresh apple's image and 2342 rotten apple's image. Hence a total of 4035 number of images has been taken to train the UNet / En-UNet model.

3.3 Network architecture

The proposed work is compiled using UNet architecture and a modified version of the same. The modified architecture is found to generate enhanced outputs as compared to UNet and hence is named as Enhance UNet. The block diagram representation of both the networks is shown in Figs. 4 and 5 respectively. The UNet architecture is a category of fully convolutional network performing down-sampling and up-sampling. The down- and up-sampling processes are connected with a concatenation operator and hence maintain symmetry of the architecture. The UNet architecture predicts a good segmentation map combining the localization and contextual information from the sampling process. In the present work the task of segmentation has been done using UNet and En-UNet. The results obtained are more promising in case of En-UNet and has been discussed in details in the result section.

3.3.1 UNet

In UNet the top-most layer being the input layer an image of size $128 \times 128 \times 3$ is fed. For each of the encoder blocks, 2 convolution layers and 1 maxpooling layer is

present and a sequential interconnection between the encoder and decoder block is retained. The decoder blocks consist of the convolution transpose layer for the expansion purpose with a concatenation layer. The maximum depth of convolution used is 128. Adam optimizer is present in the UNet architecture. Adam optimizer accelerates the search in the direction of finding out the minima, however sometimes it is missed due to the step size of the gradient. Binary cross entropy is used to calculate the loss function.

3.3.2 En-UNet

In the modified architecture of the UNet named as En-UNet, each encoder block consists of 2 convolutional 2D layer, 1 dropout layer with one maxpooling layer and the decoder block encompasses 2 convolution 2D layer, 1 dropout layer and Convolution 2D transpose layer with the concatenation. Along with these layers, between the encoder and the decoder, an extra layer of 2D convolution, maxpooling, 2D convolutional transpose layer is also used in the modified architecture. Due to the addition of these extra layers in the architecture the maximum convolutional depth is increased to 512. Hence the layers present in the enhanced UNet can be represented as: (i). Encoder: Conv 2D + Dropout + Maxpooling (ii). Decoder: Conv 2D Transpose + Concatenate + Conv2D + Dropout.

The encoder block1 as shown in Fig. 5 consists of 2 convolution layers, each with 16 filters of kernel size 3×3 followed by the activation function 'elu' (exponential linear unit). The image is then passed through a pooling layer of size 2×2 . The procedure is repeated in encoder block 2, 3 and 4 with 32 filters, 64 filters and 128 filters each. After these encoder blocks an additional layer has been incorporated which enhanced the quality of the output more than the general UNet. Incorporation of this layer

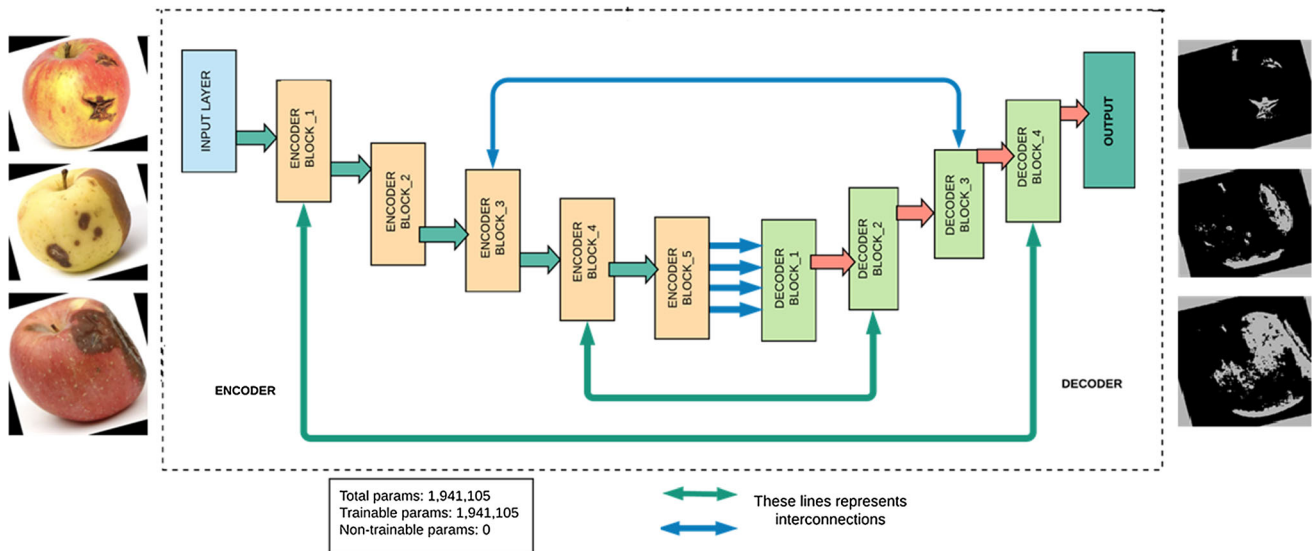


Fig. 4 Block diagrammatic representation of UNet

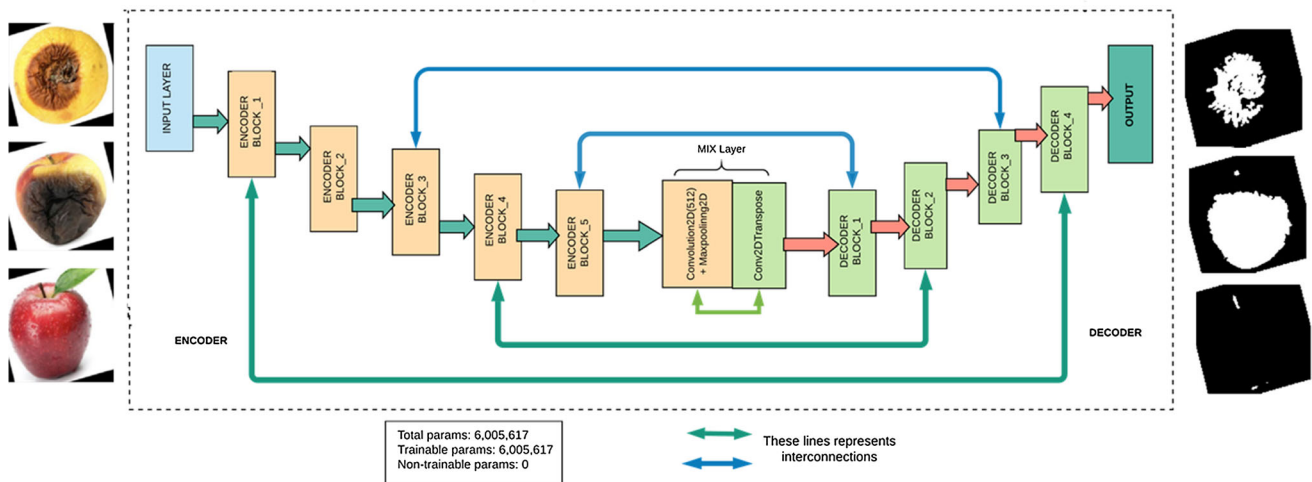


Fig. 5 Block diagrammatic representation of En-UNet

increased the depth and thereby extracts more features in the feature map producing desired outputs. This layer is a conv 2D with 512 filters of 3×3 kernel size and conv 2D transpose with 256 filters connected directly to the decoder. Each decoder block include the number of filters in the decreasing order as 128, 64, 32 to 16 for the up-sampling process. Each decoder block is concatenated with the corresponding encoder block maintaining symmetry. The output has one filter and 'sigmoid' activation function. In the output, segmentation of the rotten portion present in the input RGB apple's image is obtained in white color and the output is entirely black if the input image is that of a fresh apple. The segmented output is different for fresh and rotten apple which distinguishes the task and generates the actual prediction of being fresh / rotten from the RGB input image. The proposed model is compiled using RMS-Prop

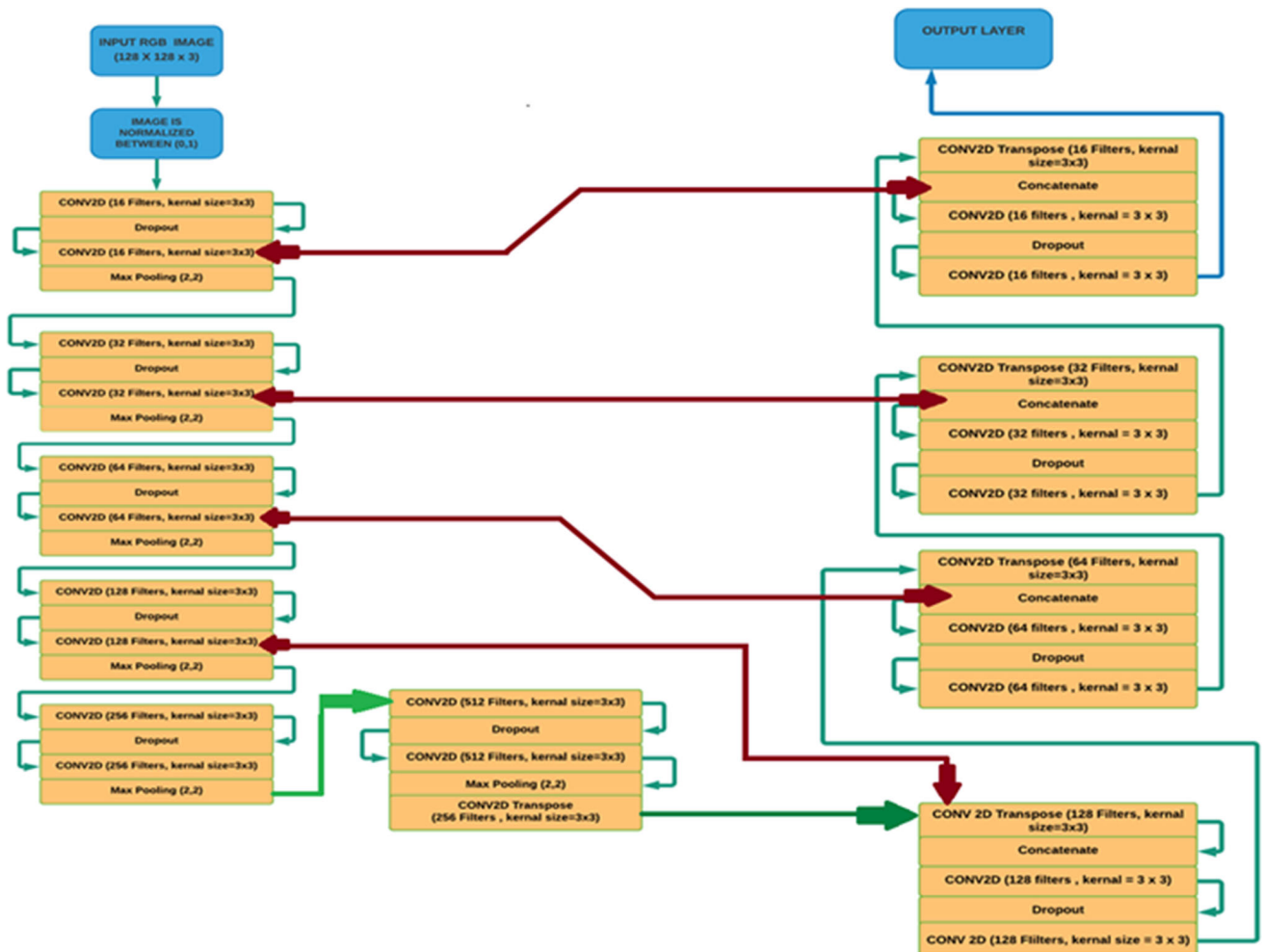
optimizer and binary cross entropy as the loss function. The total number of layers present is reported in Table 1. Figure 6 portrays the type of convolutional layers with the number of filters used in segmentation task.

3.4 Training module

Training accuracy gives the actual prediction of the input image being rotten or fresh. In the training module two types of architectures are trained viz. UNet and En-UNet; UNet being the backbone of the modified architecture (En-UNet). In the training module 3102 images of size $128 \times 128 \times 3$ are trained in 97 batches. Each batch includes 32 images. The training time consumed by UNet is observed to be 37 min while that of En-UNet is 46 min. The instant training accuracy and training loss per epoch

Table 1 Type and number of layers in UNet and En-UNet

Layers architecture	2D Conv	Drop out	2D Max Pooling	2D Conv. Transpose	Concatenation
U-NET	19	9	4	4	4
Proposed En-UNet	21	11	6	5	4

**Fig. 6** The convolutional layers of En-UNet for segmentation

have been monitored from the event log files of the Tensorflow model.

3.5 Trained module

The trained model is the outcome of the training module with the mentioned architectures. In the trained model any random input test image (rotten/fresh apple) can be fed and the result generated is the actual deciding predictions. The trained model also gives the model accuracy estimation performance of the segmentation task.

4 Results and discussion

A comprehensive study based on simulation has been accomplished and the proposed deep learning segmentation models are validated against various performance parameters. Training of the models are done in Google Colabs with GPU Tesla K80 (2496 CUDA cores).

The deep learning segmentation architectures (UNet and En-UNet) that are executed in the present work are validated against the RGB image data, where En-UNet have shown better results than UNet. The objective behind this experiment was to explore the performance of both the models on the segmentation task. The UNet achieved training and validation accuracies of 93.19% and 95.36%

respectively on segmentation of apples and encountered training and validation loss of 0.07 and 0.1086 respectively. The training and validation accuracies obtained by En-UNet are 97.46% and 97.54% respectively against training loss of 0.0657 and validation loss of 0.0618. The graphical representation of the accuracies and loss are depicted in Fig. 7. The predicted output should have a match of ground truth masks generated from the input RGB apple's images. The generated masks and corresponding predicted output for both the architectures are shown in Fig. 8. From the figure it can be observed that the predicted output finds best match with the ground truth images for En-UNet over UNet.

To evaluate the semantic segmentation model, IoU (Intersection Over Union) metric is very effective and commonly used. This IoU is also known as Jaccard Index. The IoU score of each class is calculated and then the mean IoU is computed to present a global IoU score on the semantic segmentation. The average IoU under a threshold of 0.95 with the input and predictions for the models used in the work is reported in Table 2 and the differences in the values of the mean IoU can be witnessed. The best IoU

score achieved is 0.866 with En-UNet and 0.66 with that of UNet.

The segmentation results obtained from UNet and En-UNet are represented in Fig. 9 for a comparable visualization. Better results are obtained from the En-UNet model which can also be seen from the figure. The segmented outputs for both rotten and fresh apple are displayed. For rotten apple images the segmentation got enhanced in case of En-UNet segmentation and every region under rotten portion is present in the output image. In UNet, segmentation brings out lesser portion of rotten areas as compared to En-UNet. Moreover in case of fresh apples the result shows that UNet has segmented out some portion in the fresh apple owing to over segmentation where as En-UNet has done justice to the task of segmentation and produced nearly the same output as the ground truth (black) for fresh apples. The total trainable parameters for UNet are 1,941,105 whereas those for En-UNet are 6,005,617. The optimizer RMS Prop used for En-UNet also facilitated the architecture for perfect finding of the minima.

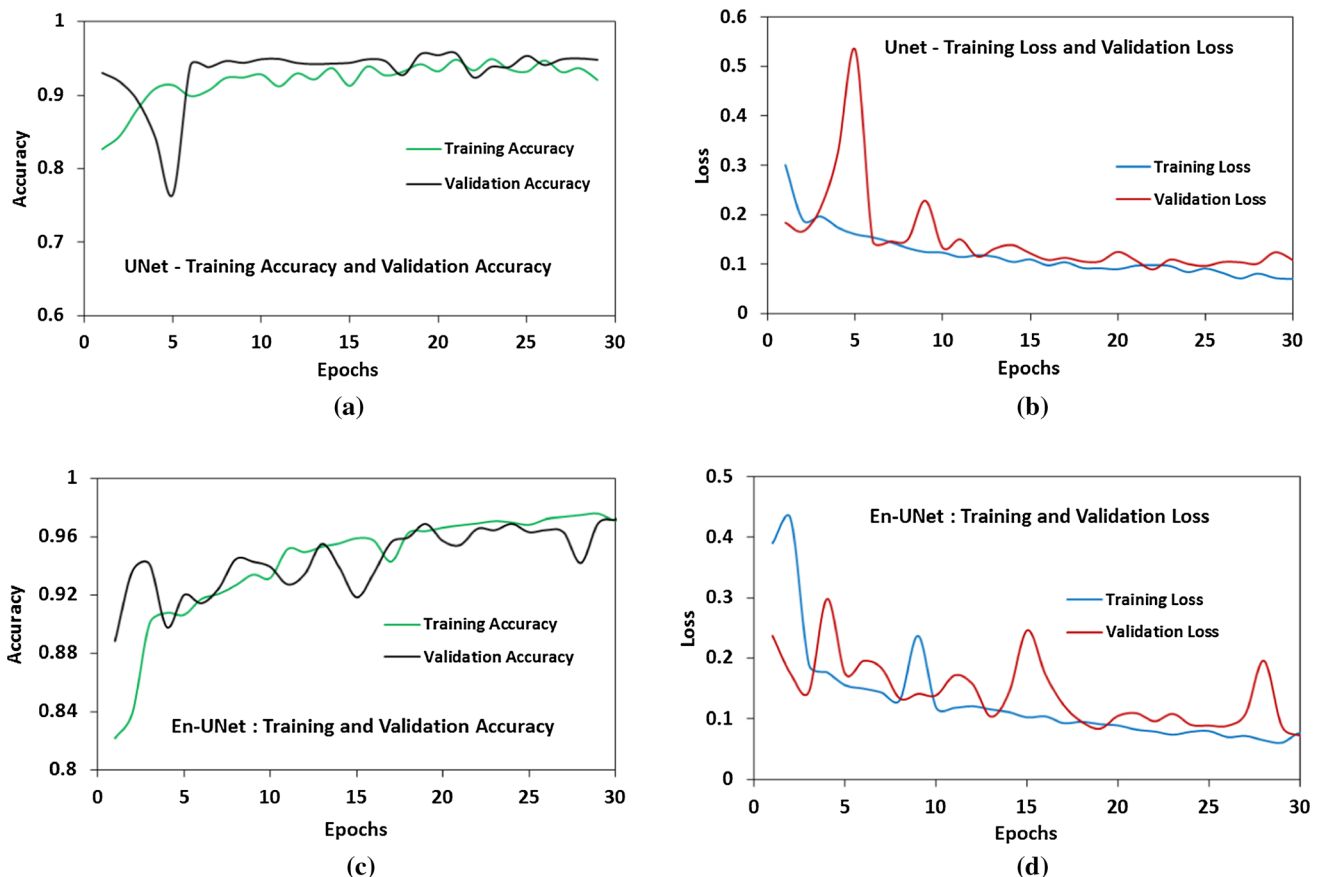


Fig. 7 Accuracy and loss plot **a** training and validation accuracy of UNet **b** training and validation loss of UNet **c** training and validation accuracy of En-UNet **d** training and validation loss of En-UNet

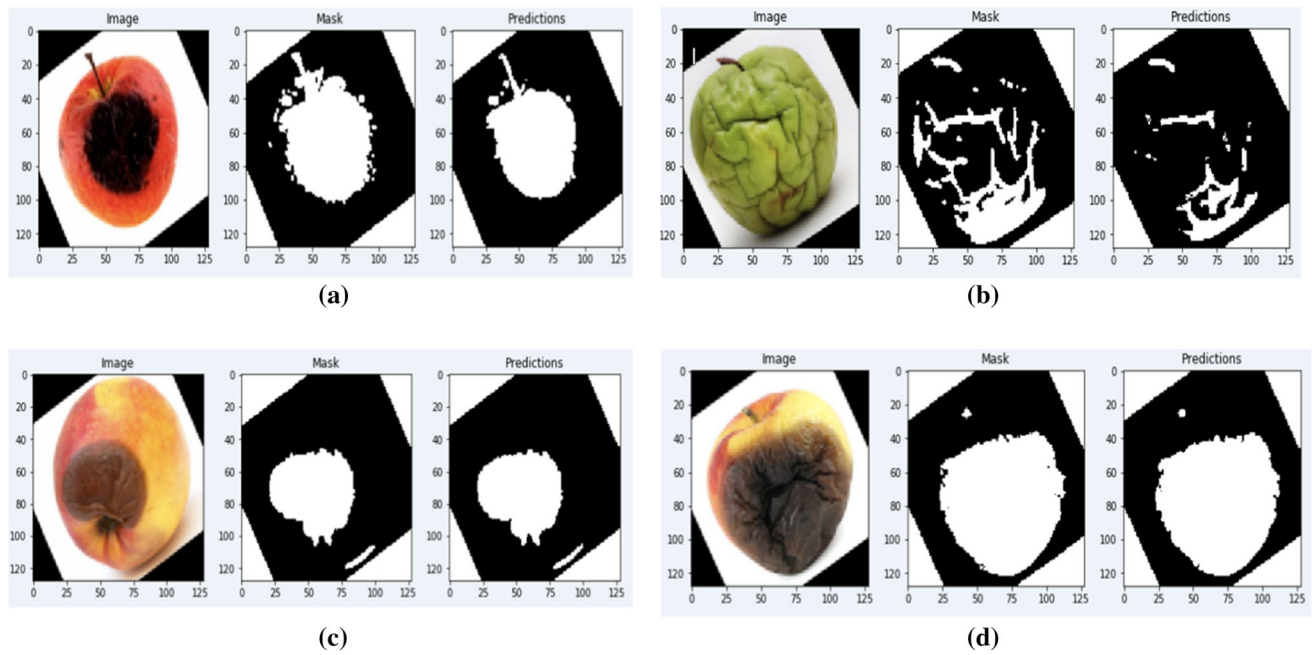
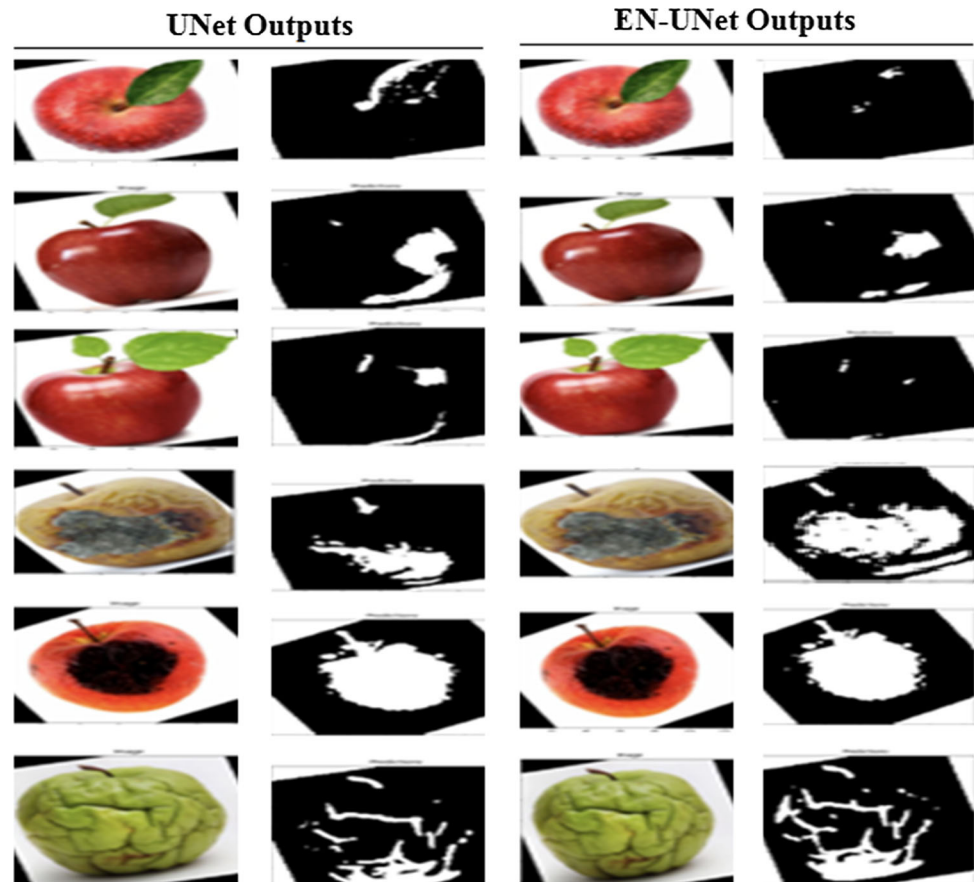


Fig. 8 RGB image mask and predicted output (a–b) UNet, (c–d) En-UNet

Table 2 Semantic segmentation with UNet and En-UNet and performance parameter—Mean IoU for different output Images (A stands for UNet and B stands for En-UNet)

Input Image with predicted Output (UNet)	Input Image with predicted Output (En-UNet)	IoU_A	IoU_B
		0.66	0.86
		0.33	0.35
		0.33	0.45
		0.21	0.27
		0.45	0.50

Fig. 9 Segmented output images of UNet and En-UNet



5 Conclusion

Real time semantic segmentation of rotten apples is carried out leading to categorization of fresh apples from the rotten ones. The RGB images of the rotten and fresh apples are fed to two architectures viz. UNet and En-UNet. Before feeding the image data to the networks, binary masks as ground truths are generated and then the predicted outputs are validated against the ground truths. Comparisons of both the models are included in the paper in terms of accuracy and IoU score. From the results, UNet achieved validation accuracy of 95.36% whereas En-UNet achieved 97.54% validation accuracy. The best mean IoU score under a threshold of 0.95 attained by En-UNet is 0.866 and that of UNet is 0.66. The ability of the proposed work can be very much useful in the horticulture domain as well as fruit industries with fast automation system developed for good quality fruit detection and production of quality food products. The present work will definitely help the researchers with new ideas for building sophisticated automation system for smart agriculture and horticulture based industries.

Compliance with ethical standards

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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